

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0518

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

16. Object Detection under Scale Variation

Results:

Object Size	Accuracy
Large	94%
Medium	80%
Small	55%

Compare detection performance and evaluate system limitations. Justify improvements.

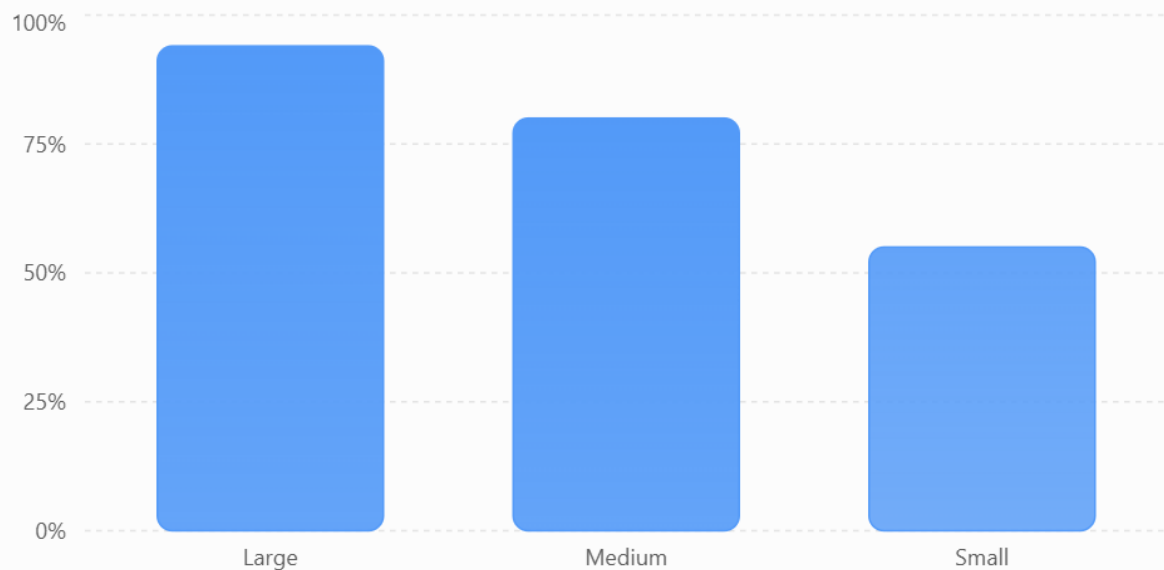
ANSWER:

Object detection accuracy by object size

Detection performance decreases as object size becomes smaller.

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Comparison:

- **Large objects – 94%:** Best detection performance because they contain more pixels and clearer features.
- **Medium objects – 80%:** Performance decreases because fewer visual details are available.
- **Small objects – 55%:** Lowest accuracy due to loss of fine details, low resolution, and difficulty distinguishing the object from the background.

System limitations:

1. The system is highly affected by object size.
2. Small objects may be missed or incorrectly classified.
3. Image resizing and down sampling can remove important object features.
4. Occlusion, complex backgrounds, and low-quality images can further reduce small-object detection accuracy.

Improvements:

- Use multi-scale object detection such as Feature Pyramid Networks (FPN).
- Increase input image resolution.
- Use data augmentation with different object sizes and scales.
- Include more small-object samples during training.
- Use better feature extraction and detection models.

- Apply multi-scale training and testing to improve robustness.

Conclusion:

The system performs well for large objects (**94%**) but has a significant limitation for small objects (**55%**). Therefore, improving multi-scale feature extraction, image resolution, and small-object training is necessary to achieve more consistent detection performance across different object sizes.

Object detection accuracy by object size

Detection performance decreases as object size becomes smaller.

size	accuracy
Large	94
Medium	80
Small	55

17. Stereo vs SFM Accuracy Comparison

Results:

Method	Accuracy	Hardware
Stereo	High	Dual camera
SFM	Moderate	Single camera

Compare and evaluate which is more practical under hardware limitations.

ANSWER:

Method	Accuracy	Hardware Required	Evaluation
Stereo Vision	High	Dual camera	More accurate depth estimation but requires two cameras with proper calibration.
SFM (Structure from Motion)	Moderate	Single camera	Lower accuracy but works with only one camera and is cheaper.

Comparison:

- Stereo Vision provides higher accuracy because two cameras capture the scene from different viewpoints, allowing accurate depth estimation.

- SFM uses a single camera and estimates 3D structure from multiple images taken from different positions.
- Stereo requires more hardware, calibration, and synchronization.
- SFM has lower hardware requirements and is easier to deploy in resource-constrained systems.

Under hardware limitations: SFM is more practical because it requires only a single camera, reducing cost and hardware complexity. Although its accuracy is moderate, it is suitable when budget, space, or hardware availability is limited.

Conclusion:

If accuracy is the priority, Stereo Vision is better. If low hardware cost and simplicity are the priorities, SFM is the more practical choice.

20. Surveillance Model Selection

Results:

Model	Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Compare and evaluate under strict latency constraints.

ANSWER:

Model	Accuracy	Latency	Evaluation
A	82%	30 ms	Fastest, but lower accuracy
B	90%	70 ms	Good balance between accuracy and speed
C	95%	120 ms	Highest accuracy but slowest

Comparison:

- **Model A** has the lowest latency (**30 ms**), making it suitable for real-time surveillance, but its accuracy is only **82%**.
- **Model B** provides **90% accuracy** with **70 ms latency**, giving a better balance.

- **Model C** has the highest accuracy (**95%**) but **120 ms latency**, which may be too slow for strict real-time surveillance.

Under strict latency constraints: Model A is the best choice if the system requires very fast response. If the latency limit allows around **70 ms**, **Model B is preferable** because it provides significantly better accuracy while maintaining relatively low latency.

Conclusion:

For very strict latency, choose Model A. For a practical balance between speed and accuracy, choose Model B. Model C should be avoided when low latency is critical.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

ANSWER:

Noise Level	Accuracy	Performance
Low	92%	Very good detection
Medium	80%	Moderate detection
High	60%	Poor detection

Comparison:

- At **low noise**, the system achieves **92% accuracy**, showing good detection performance.
- At **medium noise**, accuracy drops to **80%**, indicating that noise affects feature extraction and object recognition.
- At **high noise**, accuracy falls significantly to **60%**, showing that the system struggles to identify objects correctly.

System Robustness: The system has good robustness at low noise, but its robustness decreases as noise increases. The 32% drop in accuracy from low to high noise shows that the model is sensitive to noisy input.

Improvements:

1. Apply image denoising and filtering before detection.
2. Train the model using noisy and clean images.
3. Use data augmentation with different noise levels.
4. Improve feature extraction to make the model less sensitive to noise.

Conclusion:

The system performs well under low noise but is not highly robust to severe noise. Noise reduction and training with noisy images can improve detection reliability.

23. Multi-Model Comparison

Results:

Model	Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High
C	95%	Slow	High

Compare and evaluate the best overall model.

ANSWER:

Model	Accuracy	Speed	Robustness	Evaluation
A	88%	Fast	Low	Good for real-time applications but less reliable
B	92%	Moderate	High	Good balance of accuracy, speed, and robustness
C	95%	Slow	High	Highest accuracy and robust, but slower

Comparison:

- **Model A** is the fastest, but its **low robustness** makes it less reliable in changing conditions.
- **Model B** provides **92% accuracy**, moderate speed, and high robustness, giving a strong overall balance.
- **Model C** has the highest accuracy (**95%**) and high robustness, but its slow speed can be a limitation for real-time applications.

Best Overall Model: Model B

Model **B** is the **best overall choice** because it provides a good balance between **accuracy (92%)**, **speed**, and **robustness**. Although Model C has higher accuracy, its slow processing makes it less practical for real-time systems.

Conclusion:

Model B is the most practical overall model, while Model C is preferable when maximum accuracy is more important than speed.

26. Feature Detection under Illumination Changes

Results:

Method	Normal	Low Light
A	90%	60%
B	85%	75%

Compare and evaluate robustness.

ANSWER:

Method	Normal Light	Low Light	Accuracy Drop
A	90%	60%	30%
B	85%	75%	10%

Comparison:

- **Method A:** Higher accuracy in normal lighting (**90%**), but performance drops to **60%** in low light.
- **Method B:** Slightly lower accuracy in normal lighting (**85%**), but maintains **75%** accuracy in low light.
- Method A has a **30% accuracy drop**, while Method B has only a **10% drop**.

Robustness Evaluation: Method B is more robust because it maintains better performance when lighting conditions change.

Conclusion:

Method A is better in normal lighting, but Method B is the better overall choice for environments with variable or low illumination because its performance is more stable.

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Compare and evaluate suitability.

ANSWER:

Pipeline	Accuracy	Speed	Suitability
A	88%	Fast	Best for real-time applications
B	92%	Moderate	Best overall balance
C	95%	Slow	Best when accuracy is more important than speed

Comparison:

- **Pipeline A** has the fastest processing speed but lower accuracy (**88%**). It is suitable for real-time video applications.
- **Pipeline B** provides **92% accuracy** with moderate speed, giving a good balance between performance and processing time.
- **Pipeline C** achieves the highest accuracy (**95%**) but is slow, making it less suitable for real-time systems.

Evaluation:

- For **real-time applications** → **Pipeline A**
- For **balanced performance** → **Pipeline B**
- For **maximum accuracy** → **Pipeline C**

Conclusion:

Pipeline B is the most suitable overall because it provides a strong balance between accuracy and speed. If the application has very strict real-time requirements, Pipeline A is preferable.

29. Detection under Occlusion

Results:

Method	Accuracy
A	70%
B	85%
C	90%

Compare and evaluate robustness under occlusion.

ANSWER:

Method	Accuracy	Evaluation
A	70%	Poor performance under occlusion
B	85%	Good performance
C	90%	Best performance

Comparison:

- **Method A** achieves only **70% accuracy**, showing that it struggles when objects are partially hidden.
- **Method B** achieves **85% accuracy**, indicating better handling of occluded objects.
- **Method C** achieves the highest accuracy of **90%**, making it the most effective under occlusion.

Robustness Evaluation: Method C is the most robust because it maintains the highest detection accuracy when objects are partially blocked or hidden.

Conclusion:

Method C is the best choice for environments with frequent occlusion, followed by Method B. Method A is less reliable because its detection accuracy is significantly lower.

Code of Conduct:

*I _HARITHI S (192421151)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

HARITHI S

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand